SURVEY DATA CLEANING AND MANAGEMENT

Guidelines & Process Document

A Step-by-Step Guide for Research Teams and Data Staff

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1. Version History and Change Log

Always check this table first to confirm you are reading the most current version of this guideline. When a new version is released, all staff must be notified and must read the updated document before continuing to work with data.

Version	Date	Author	Summary of Changes	Approved By
1.0	April 2026	Data Management Unit	Initial release - full guideline covering all 17 sections.	Principal Investigator

> **NOTE:** *When submitting a new version, update the header of this document, notify all team members by email, and archive the previous version in the /Archive/ folder with its version number in the file name.*

2. Purpose and Scope

This document explains how to clean and manage survey data in a careful, step-by-step way. Its goal is to make sure that the data your team collects is accurate, complete, and ready to use for analysis.

Data cleaning is the process of finding and fixing errors, removing extra or repeated information, and organizing data so it can be trusted. Without proper cleaning, mistakes in the data can lead to wrong conclusions - even when the rest of the research is done correctly.

This guideline is for everyone who works with survey data, including data entry staff, research assistants, and team leaders. It covers the full journey of data - from the moment it is collected to when it is ready for final analysis.

2.1 Why Data Cleaning Matters

Think of raw survey data like fresh vegetables from a farm. Before you can cook with them, you need to wash them, remove any bad pieces, and cut them the right way. Data cleaning works the same way. It prepares your data so it is safe and useful.

Poor data quality can cause these problems:

- Wrong numbers that change the results of your study
- Missing answers that make it hard to compare groups
- Repeated records that make your sample look bigger than it is
- Confusing labels that lead to misunderstandings

2.2 Scope of This Document

This guideline covers survey data that is collected through these methods:

- Paper-based questionnaires that are entered into a computer by hand
- Mobile data collection using tools like KoBoToolbox, ODK, or SurveyCTO
- Online surveys using platforms like Google Forms or SurveyMonkey
- Interview-based surveys where an enumerator fills out the form

> **NOTE:** *This document does not cover qualitative data (like focus group notes or open-ended interview transcripts) or administrative data from government records.*

3. Key Terms and Definitions

Before you begin cleaning data, it helps to understand the basic words used in this process. The table below explains the most important terms in plain language.

Term	What It Means
Raw Data	Data exactly as it was collected - before any checking or corrections have been made.

Term	What It Means
Clean Data	Data that has been checked, corrected, and organized so it is ready for analysis.
Missing Value	A blank answer in the dataset - where no response was given or recorded.
Outlier	A value that is very different from all the other values. It may be a real answer or a mistake.
Duplicate	A record that appears more than once - meaning the same person or household was counted twice.
Codebook	A reference document that explains what each variable name and code number means in the dataset.
Variable	A column in your dataset, representing one question or piece of information collected from each person.
Respondent	The person who answered the survey questions.
Data Entry Error	A mistake made when typing data into the computer - for example, typing 999 instead of 99.
Skip Logic	A survey rule that tells respondents to skip certain questions based on earlier answers.

4. Roles and Responsibilities

Good data cleaning does not happen by one person alone. It requires a team where everyone knows their job. The table below shows the main roles involved in survey data management and what each role is responsible for.

Role	Key Responsibilities	When They Are Involved
Data Manager	Oversees the whole cleaning process; makes final decisions on problem records	All stages
Data Entry Clerk	Types paper forms into the system; flags unclear answers	Data entry stage
Field Supervisor	Checks completed forms before they leave the field; follows up with enumerators	Data collection stage
Research Analyst	Runs checks and frequency tables; identifies patterns of error	Cleaning and analysis stage
Principal Investigator	Approves changes to the dataset; reviews the final clean file	Final review stage

5. Training and Onboarding Guidance for New Staff

Every new team member who will handle survey data must complete an onboarding process before they work on any live dataset. This section explains what that process looks like and what each new staff member needs to know.

5.1 Before Starting Work

New staff must complete all of the following before they are given access to any project data:

1. Read this entire guideline document and sign the confirmation form at the end.
2. Attend the data cleaning orientation session run by the Data Manager (minimum 2 hours).
3. Complete the practice exercise using the sample dataset provided by the team (see Section 5.3 below).
4. Have their practice exercise reviewed and approved by the Data Manager.
5. Sign the data confidentiality agreement kept on file by the organization.

5.2 Key Rules Every New Staff Member Must Know

1	Never work on the original raw data file. Always use a copy.
2	Every change to the data must be recorded in the Data Cleaning Log - no exceptions.
3	When in doubt about any value, flag it and ask the Data Manager. Never guess.
4	Use only the agreed missing value codes from Section 7. Never leave a cell blank without a code.
5	Do not share data files with anyone outside the team without written permission.
6	If you make a mistake, report it immediately. Mistakes caught early are much easier to fix.
7	Follow the file naming and storage rules in Section 13. Do not save files to your personal desktop.

5.3 Practice Exercise

Before working on real project data, every new staff member completes a practice exercise. The exercise uses a fake dataset with 50 records that contains deliberately planted errors - including duplicates, out-of-range values, skip logic mistakes, and missing values.

The new staff member must identify and correct all errors, fill out a Data Cleaning Log for each change, and produce a clean version of the practice dataset. The Data Manager reviews this work and provides feedback before granting access to live data.

> **NOTE:** *The practice dataset and answer key are stored in the shared team folder under: /Training/DataCleaning/PracticeDataset/*

5.4 Ongoing Learning

- All data staff attend a quarterly team review where common mistakes are shared and discussed (no names attached).
- When new tools or procedures are introduced, a short briefing session is held before they are used on live data.
- New versions of this guideline are shared with all staff as soon as they are released. Staff must confirm they have read updates.

6. Data Privacy and Ethical Handling

Survey data often contains sensitive personal information about respondents - their age, income, health, location, and more. Protecting this information is not only a good practice; it is an ethical responsibility and often a legal requirement.

6.1 Anonymization

Anonymization means removing or hiding any information that could identify a specific person. Before the clean dataset is shared outside the core data team, the following steps must be taken:

- Remove or hide names, phone numbers, national ID numbers, and GPS coordinates
- Replace unique identifiers with anonymous codes (e.g., replace 'John Doe' with 'HH0042')
- Consider geographic aggregation - for example, only show district-level location, not village or GPS point

6.2 Access Control

- Only authorized team members may access the data files
- Password-protect all data files stored on shared drives
- Use separate 'analysis' versions of the data that do not include personally identifiable information
- Keep a record of who accessed the dataset and when

6.3 Data Sharing

When sharing data outside the organization - for example, with a partner, donor, or publication - always check the research ethics agreement or data-sharing protocol first. Never share data that has not been formally approved for sharing.

> **NOTE:** *If your project follows specific data protection laws - such as the Bangladesh Data Protection Act or GDPR - make sure your team has read and follows those rules. When in doubt, ask your organization's ethics officer or legal team.*

7. Glossary of Error Codes and Missing Value Codes

All team members must use the same codes for missing or problematic values. Never leave a cell blank without a code - a blank cell could mean 'zero', 'skipped', 'refused', or 'not asked', and these are all very different things. Use the codes below consistently across your entire dataset.

7.1 Standard Missing Value Codes

Code	Label	When to Use It	Example
-98	**Refused**	Respondent was asked but chose not to answer	*Refused to give income amount*
-99	**Don't Know**	Respondent genuinely did not know the answer	*Did not know household size*
-95	**Not Applicable**	Question did not apply due to skip logic	*No phone model asked because no phone owned*
-66	**Not Asked**	Enumerator did not ask the question (interviewer error)	*Question skipped by mistake*
-55	**Illegible**	Answer was written on the paper form but cannot be read	*Handwriting unreadable on age question*
-44	**Data Entry Error**	A value was recorded but is clearly impossible or contradictory	*Age entered as 999*
-11	**Missing - Unknown**	Cell is blank and the reason cannot be determined	*No value in column, no paper form available*

7.2 Common Error Flag Codes

During cleaning, use these short codes in a separate 'flag' column to mark records that need attention. Do not delete the original value - just add the flag code next to it.

Flag Code	Flag Name	Description
FLAG_OOR	Out of Range	Value is outside the expected minimum or maximum for this variable
FLAG_DUP	Duplicate Record	This record appears to be a duplicate of another respondent ID
FLAG_SKIP	Skip Logic Error	An answer exists where the question should have been skipped, or vice versa
FLAG_INCON	Inconsistent	Two answers in the same form contradict each other
FLAG_OUTL	Outlier	Value is statistically extreme - check if real or an error
FLAG_REVIEW	Needs Review	Record needs manual review by the Data Manager before a decision is made

8. Data Cleaning Process - Visual Flowchart

The diagram below shows the full data cleaning process from start to finish. Use it as a quick reference during your work. Detailed instructions for each step are in Section 9.

STEP 1 · COLLECT RAW DATA
Paper forms, mobile devices, or online platforms

▼

STEP 2 · BACK UP RAW FILE
Save original; never edit the master copy

▼

STEP 3 · IMPORT & COUNT RECORDS
Verify row count matches field log

▼

◆ Does record count match the field log?
✓ YES: Continue to Step 4 | ✗ NO: Investigate discrepancy before continuing

▼

STEP 4 · CHECK COMPLETENESS
Find all missing values; decide on coding

▼

STEP 5 · REMOVE DUPLICATES
Use unique ID to find and remove repeat records

▼

STEP 6 · CHECK VALUE RANGES

Flag out-of-range numbers and correct or recode

▼

◆ **Are errors > 5% of records?**
✓ YES: Stop - re-clean the full dataset | ✗ NO: Continue to next step

▼

STEP 7 · CHECK SKIP LOGIC
Ensure filter questions match follow-up answers

▼

STEP 8 · STANDARDIZE FORMATS
Dates, codes, text case - one consistent format

▼

STEP 9 · HANDLE OUTLIERS
Review each outlier; document every decision

▼

STEP 10 · DOCUMENT ALL CHANGES
Complete the Data Cleaning Log - no exceptions

▼

STEP 11 · PEER REVIEW & SPOT-CHECK
Second team member checks 10% of records

▼

◆ **Does spot-check pass (errors < 5%)?**
✓ YES: Continue to final step | ✗ NO: Return to Step 4 and re-clean

▼

STEP 12 · SAVE & LOCK CLEAN FILE
Final clean file saved, named, and locked

9. The Data Cleaning Process - Step by Step

Follow these steps in order. Do not skip any step. Each step builds on the one before it. If you find a serious problem at any step, stop and report it to the Data Manager before continuing.

#	Step Name	What You Do
1	**Back Up the Raw Data**	Before you touch anything, copy the original dataset and store it in a safe folder. Name the file clearly - for example: 'ProjectName_RawData_YYYYMMDD'. Never work directly on the original file.
2	**Import and Review**	Open the dataset and look at its basic shape. Count the number of rows (respondents) and columns (variables). Check that the numbers match what was expected from the field.
3	**Check Completeness**	Find all questions that should not be blank. Count missing values for each variable. Decide whether missing values are acceptable or need to be followed up.

#	Step Name	What You Do
4	**Remove Duplicates**	Use the unique respondent ID to find any records that appear more than once. Review each duplicate carefully before deleting it.
5	**Check Value Ranges**	For every numeric variable, check the minimum and maximum values. Flag any values that fall outside the expected range - such as an age of 200 or a negative income.
6	**Check Skip Logic**	Compare answers to filter questions with follow-up answers. For example, if a person said they do not own a phone, the next phone-related question should be blank.
7	**Standardize Formats**	Make sure dates, codes, and categories are written the same way across the whole dataset. Convert all text to the correct case. Recode any free-text answers into proper codes.
8	**Handle Outliers**	Review flagged outlier values one by one. Decide if each is a real value or a data entry error. Document every decision you make about an outlier.
9	**Document All Changes**	Write down every correction made to the data in the Data Cleaning Log. Include the variable name, the old value, the new value, and the reason for the change.
10	**Create the Clean File**	Save a final cleaned version of the dataset with a clear name - for example: 'ProjectName_CleanData_YYYYMMDD'. Lock the file so it cannot be accidentally changed.

10. Common Data Problems and How to Fix Them

The table below describes the most common problems found in survey data and gives you clear guidance on how to deal with each one. When you are not sure what to do, always ask the Data Manager.

Issue Type	What It Means	How to Fix It
Missing Values	The respondent did not answer the question, or the answer was not recorded during data entry.	Check the original paper form or recording. If the answer cannot be found, use the project's agreed missing-value code from Section 7. Document the decision.
Out-of-Range Values	A number is too high or too low to be real. For example: Age = 0 or Age = 150.	Check the original form. Correct if a clear entry error is found. If the original answer is also wrong, recode as missing and document the reason.
Inconsistent Answers	Two answers in the same form contradict each other. For example: the respondent is listed as 5 years old but is also marked as married.	Flag the record and check the original form. If the contradiction cannot be resolved, mark the conflicting answers as missing and record a note.
Duplicate Records	The same respondent ID appears more than once in the dataset.	Compare all duplicate rows carefully. Keep the most complete and correct version. Delete the other row and document the action in the log.
Wrong Skip Logic	A question that should have been skipped has a value, or a question that should have an answer is blank.	Check the questionnaire design and the actual form. Remove values where they should not exist. Follow up to recover missing values where possible.

Issue Type	What It Means	How to Fix It
Coding Errors	Answer options are coded with the wrong number - for example, using 3 instead of 2 for 'Agree'.	Review the codebook and recode the wrong values. Check the full variable for the same mistake. Document the correction.
Format Inconsistencies	Dates or text are written in different formats - for example: '01/04/2024', 'April 1, 2024', and '2024-04-01' all in the same column.	Convert all values to one standard format agreed upon by the team. Use tools like STATA, R, or Excel formulas to do this in bulk.

11. Data Cleaning Log - Fillable Template

Use this template to record every change made to the dataset. Print additional copies or replicate this table in your spreadsheet software. One row = one change. Do not combine multiple changes in a single row.

> **NOTE:** *File this log together with the clean dataset. Both files must be stored in the same project folder and submitted together at the end of the project.*

Date	Variable Name	Resp. ID	Old Value	New Value	Reason for Change	Initials
2026-04-10	*hh_age*	*HH0042*	*250*	*25*	*Likely data entry error; original form shows '25'*	*R.K.*
2026-04-11	*income_monthly*	*HH0109*	*0*	*-99*	*Respondent refused; coded as missing per protocol*	*S.A.*

12. Quality Control and Review

Quality control (QC) means checking your own work - and having others check it too. No single person can catch every mistake. Building review steps into the process makes your data much more reliable.

12.1 Levels of Review

6. Self-check - The person who does the cleaning reviews their own work using the checklist below before passing it on.
7. Peer review - A second team member reviews a random sample of the changes in the Data Cleaning Log and checks them against the dataset.
8. Manager sign-off - The Data Manager does a final check of the complete clean dataset and approves it before it is used for analysis.

12.2 Data Cleaning Checklist

Use this checklist to confirm that all steps have been completed. The Data Manager must sign off before the dataset is shared with the analysis team.

#	Task	Responsible	When to Complete
1	Original raw data file backed up and stored safely	Data Manager	Before any cleaning begins
2	Total number of records matches the field log	Data Manager	Day 1 of cleaning
3	All variables reviewed against the codebook	Research Analyst	Day 1 of cleaning
4	Missing values identified and coded correctly	Research Analyst	Days 1-2
5	Duplicate records identified, reviewed, and removed	Research Analyst	Day 2
6	Out-of-range values checked and corrected or coded as missing	Research Analyst	Days 2-3
7	Skip logic errors identified and resolved	Research Analyst	Day 3
8	All dates and codes standardized across the dataset	Data Entry Clerk	Days 3-4
9	Outliers reviewed and documented with decisions recorded	Research Analyst	Day 4
10	Data Cleaning Log completed with all changes	All staff	Ongoing, then finalized
11	Clean dataset saved with correct naming convention	Data Manager	Final day of cleaning
12	Final clean file reviewed and approved by Principal Investigator	PI / Data Manager	Before sharing file

12.3 Spot-Checking

Spot-checking means selecting a random sample of records and checking them manually. Your team should spot-check at least 10% of all records after cleaning. If more than 5% of checked records have errors, the full dataset should be re-cleaned.

12.4 Frequency Tables

Run frequency tables for every categorical variable and minimum/maximum/mean for every numeric variable. Look for unusual patterns - one answer appearing far too often, or an unexpected option showing up in a column. Save your frequency tables as part of the project documentation.

13. File Naming and Storage Rules

Good file management keeps your data safe and makes it easy for team members to find the right version. Always follow these rules when saving data files.

13.1 File Naming Convention

Use this format for all data files:

[ProjectCode]_[DataType]_[YYYYMMDD].[ext]

Example: HHSURVEY_RawData_20260401.xlsx | HHSURVEY_CleanData_20260415.xlsx

13.2 Storage and Access Rules

- Store all project data in a designated secure folder - not on personal desktop drives.
- Use at least two storage locations: a main location and a backup (cloud + local drive).
- Only team members with project access may open or edit data files.
- Never share raw or clean data files over email or messaging apps without encryption.
- At the end of the project, archive all files and store them for at least five years.

13.3 Version Control

Never overwrite a previous file. Always save a new version with the updated date in the file name. Keep a short log of what changed between versions in a separate ReadMe file stored in the same folder.

> **NOTE:** *If two people need to edit the dataset, only one person should edit the master file at a time. Use a shared sign-out log to coordinate who is working on the file.*

14. Recommended Tools

Different tools can be used for data cleaning. The best tool for your team depends on the size of your dataset, the skills of your staff, and your organization's resources. Below are the most commonly used options.

Tool	Best For	Skill Level Needed	Cost
Microsoft Excel	Small datasets; basic checks and formatting	Beginner	Paid (often available)
Google Sheets	Small datasets; team collaboration in real time	Beginner	Free
STATA	Medium to large datasets; advanced statistical cleaning	Intermediate	Paid
R / RStudio	Any size dataset; flexible and powerful; creates cleaning scripts	Intermediate–Advanced	Free (open-source)
Python (pandas)	Large datasets; automated cleaning pipelines	Advanced	Free (open-source)

This section answers the most common questions that come up during data cleaning. If your question is not listed here, contact the Data Manager.

Q1. I found a blank cell but I am not sure why it is blank. What should I do?

Do not guess. First, check the original paper form or field recording. If you can find the reason (e.g., the respondent refused, or the question did not apply), use the correct missing value code from Section 7. If you cannot find the reason, use the code -11 (Missing - Reason Unknown) and add a note in the Data Cleaning Log.

Q2. A respondent's age looks wrong - it says 5, but they are listed as the household head. Should I change it?

Flag it as FLAG_INCON and check the original form. If the form also says 5, do not change the age - instead, flag the record for the Data Manager to review. If the form clearly shows a different number (e.g., 50), correct it, document the change, and note the reason in the log.

Q3. I found two records with the same respondent ID but slightly different answers. Which one do I keep?

Compare both records carefully. Keep the more complete and more consistent version. If both are equally complete, keep the one collected first according to the field dates. Delete the other record and document the decision in the cleaning log. If you cannot decide, escalate to the Data Manager.

Q4. My frequency table shows that one answer code appears far more often than all the others. Is that a problem?

It may or may not be. First, check if the question is one where a dominant answer is expected (for example, most people in a rural survey may answer 'farmer' for occupation). If the pattern does not make sense for the study population, investigate whether enumerators defaulted to one answer and flag the variable for review.

Q5. A respondent answered a question they should have skipped because of the filter. Should I delete their answer?

Yes - if the skip logic rule is clear and the question should not have been answered, recode the value as -77 (Not Applicable) and record the change in the log. Never just delete the value without a code. If you are unsure whether the skip rule applies, check the questionnaire and ask the Data Manager.

Q6. Someone gave an income value of 0. Is this an error or a real answer?

Zero income is possible - for example, for a student or an unemployed person. Check the questionnaire context. If adjacent answers confirm the respondent has no income, keep the zero. If it seems inconsistent (e.g., the person listed expensive assets), flag it for review.

Q7. I accidentally saved over the raw data file. What should I do?

Stop all work and report this to the Data Manager immediately. If a backup was made at the start of cleaning (as required in Step 1), the backup copy becomes the new working raw file. This is exactly why backups are mandatory. Document the incident in the project log.

Q8. Can I clean data and run analysis at the same time?

No. Analysis should only begin once the clean dataset has been formally approved and locked. Running analysis on partially cleaned data produces results that may need to be re-done later, wasting time and potentially causing confusion if two different versions of results are circulated.

16. References and Further Reading

The following resources were used to develop this guideline and can provide additional guidance for teams that want to learn more:

- World Health Organization (WHO). (2015). Data Quality Review: A Toolkit for Facility Data Quality Assessment. WHO Press.
- MEASURE Evaluation. (2019). Data Quality Assurance Tools and Guidance. Chapel Hill, NC: MEASURE Evaluation.
- UNHCR. (2021). Data Quality Monitoring Guidance for Household Surveys. UNHCR Data and Analytics Service.
- Wickham, H., & Grolemund, G. (2017). R for Data Science. O'Reilly Media.
- Inter-Agency Working Group on Data Quality (2020). Minimum Standards for Data Quality in Humanitarian Response.

17. Approval and Sign-Off

This document must be reviewed and approved before use. All team members who are responsible for data cleaning should read and confirm they understand this guideline by signing below.

Name and Title	Signature	Date

- End of Document -